

Bitwise differential trail through **LBlock** found by SAT

CiVerLy

July 24, 2026

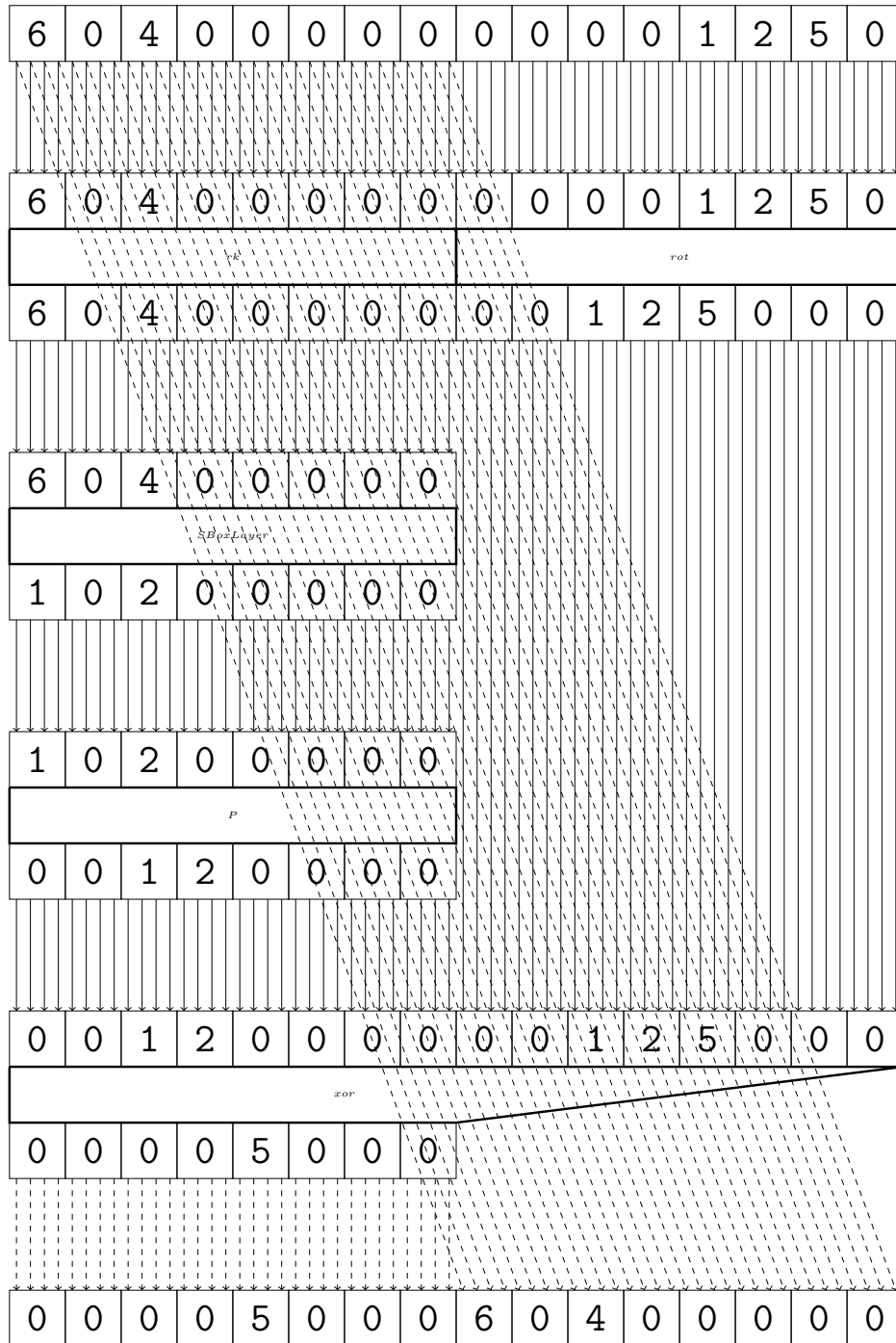
1 LBlock

Maximal differential probability: 2^{-16}

6	0	4	0	0	0	0	0	0	0	0	0	0	1	2	5	0
6	0	4	0	0	0	0	0	0	0	0	0	0	1	2	5	0
<i>LBlockRound</i>																
0	0	0	0	5	0	0	0	6	0	4	0	0	0	0	0	0
0	0	0	0	5	0	0	0	6	0	4	0	0	0	0	0	0
<i>LBlockRound</i>																
4	0	0	0	0	0	0	0	0	0	0	0	0	5	0	0	0
4	0	0	0	0	0	0	0	0	0	0	0	0	5	0	0	0
<i>LBlockRound</i>																
0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0
<i>LBlockRound</i>																
0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	0	0
<i>LBlockRound</i>																
0	0	0	0	0	0	0	5	0	0	0	0	0	0	0	4	0
0	0	0	0	0	0	0	5	0	0	0	0	0	0	0	4	0
<i>LBlockRound</i>																
0	0	0	0	4	a	0	0	0	0	0	0	0	0	0	0	5
0	0	0	0	4	a	0	0	0	0	0	0	0	0	0	0	5
<i>LBlockRound</i>																
0	0	0	0	2	5	8	0	0	0	0	0	4	a	0	0	
0	0	0	0	2	5	8	0	0	0	0	0	4	a	0	0	
<i>Final Swap</i>																
0	0	0	0	4	a	0	0	0	0	0	0	2	5	8	0	
0	0	0	0	4	a	0	0	0	0	0	0	2	5	8	0	

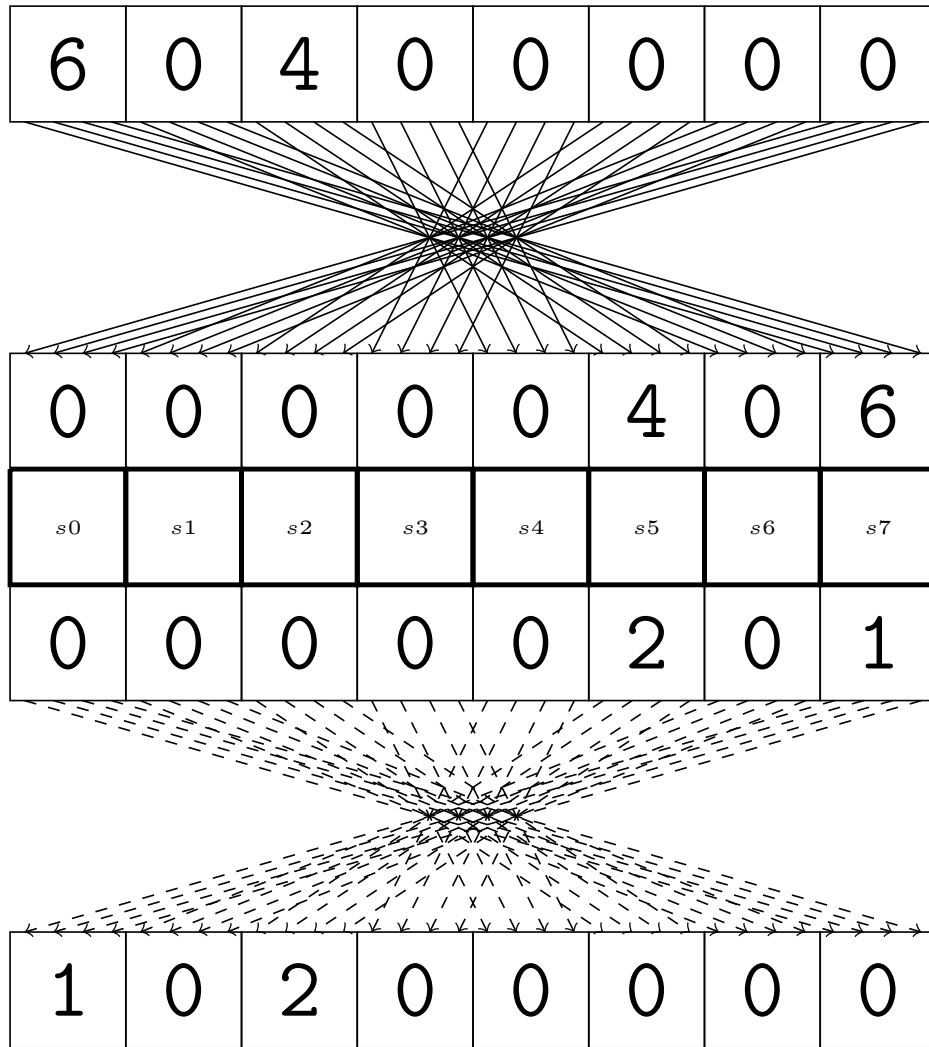
2 LBlock_round

Maximal differential probability: $2^{-4.0}$



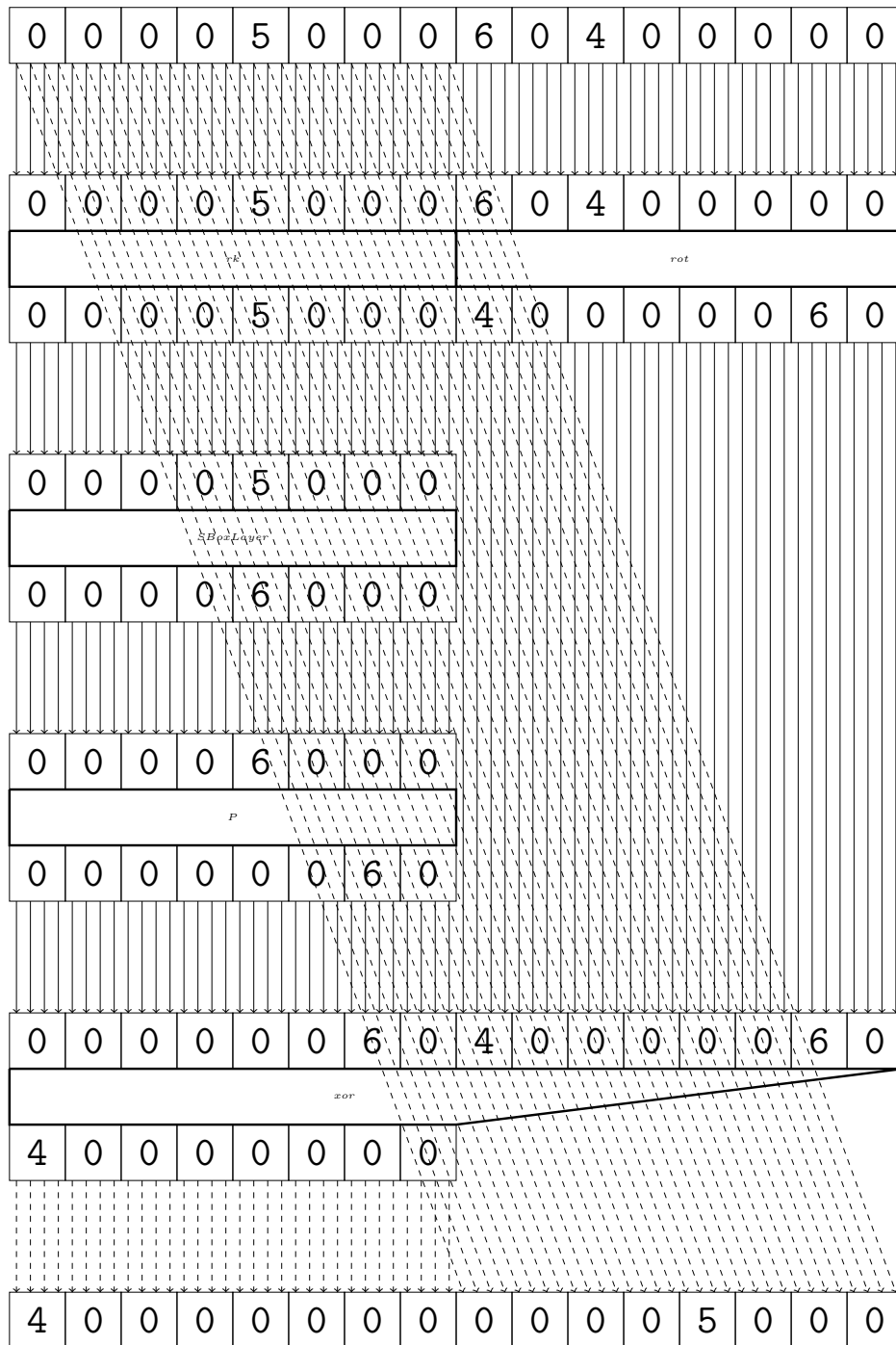
3 SBoxLayer

Maximal differential probability: $2^{-4.0}$



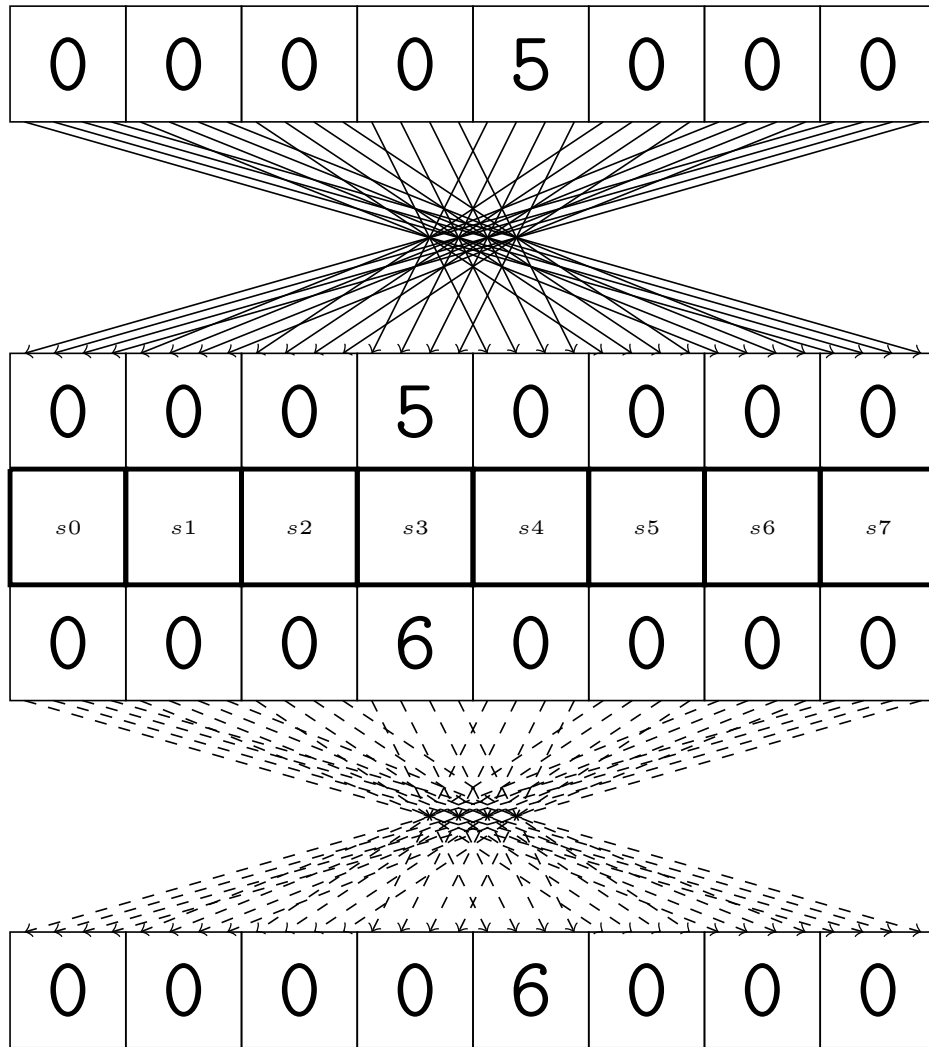
4 LBlock_round

Maximal differential probability: $2^{-2.0}$



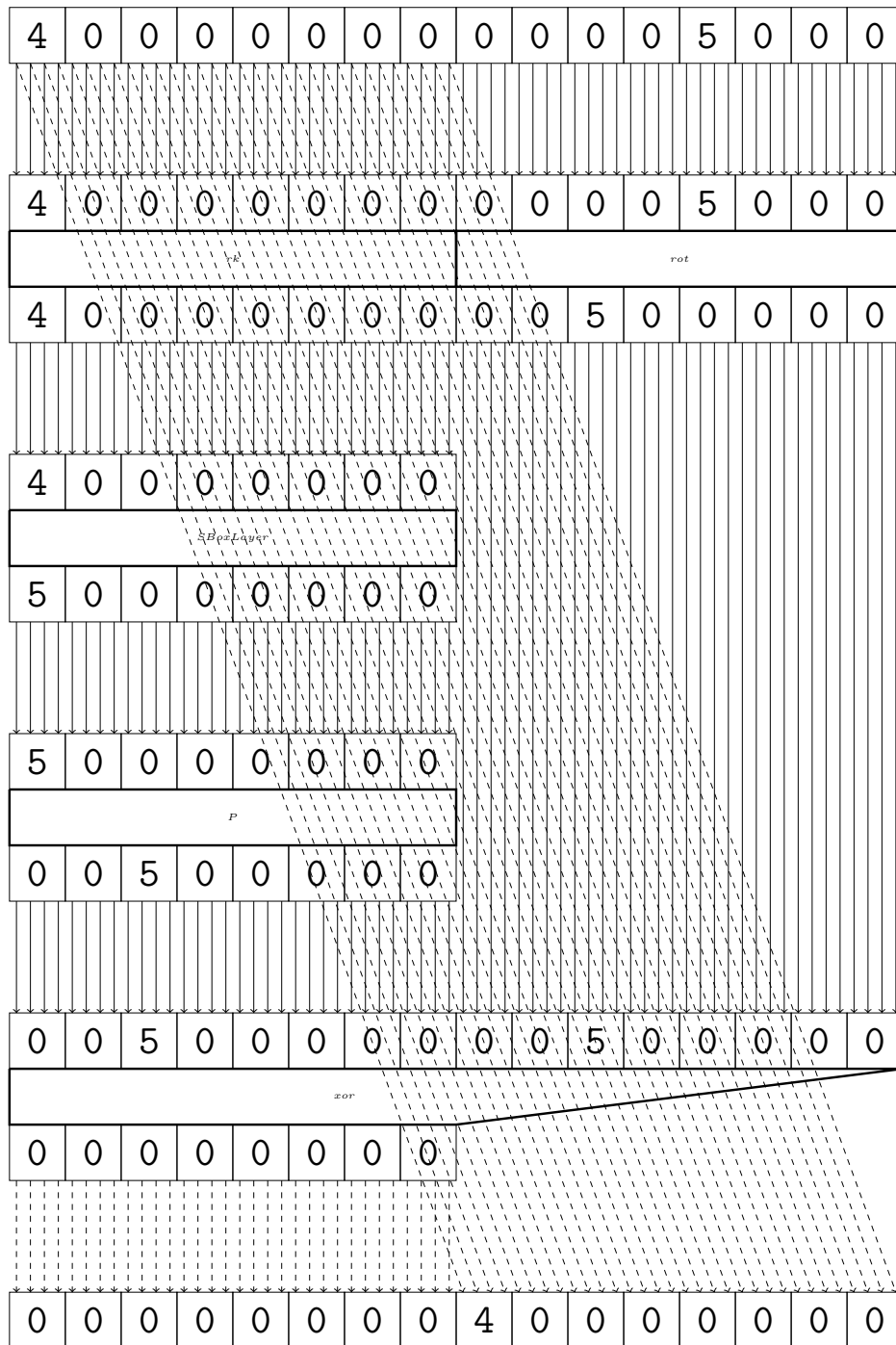
5 SBoxLayer

Maximal differential probability: $2^{-2.0}$



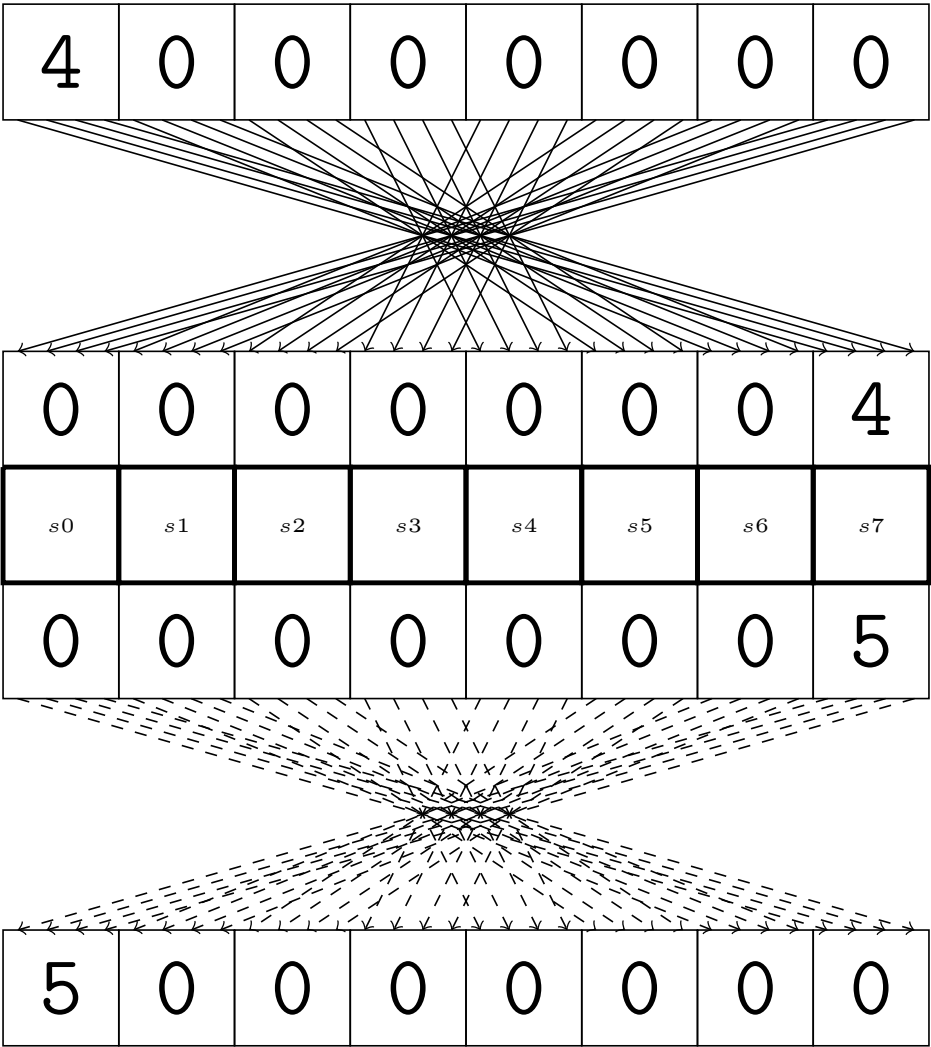
6 LBlock_round

Maximal differential probability: $2^{-2.0}$



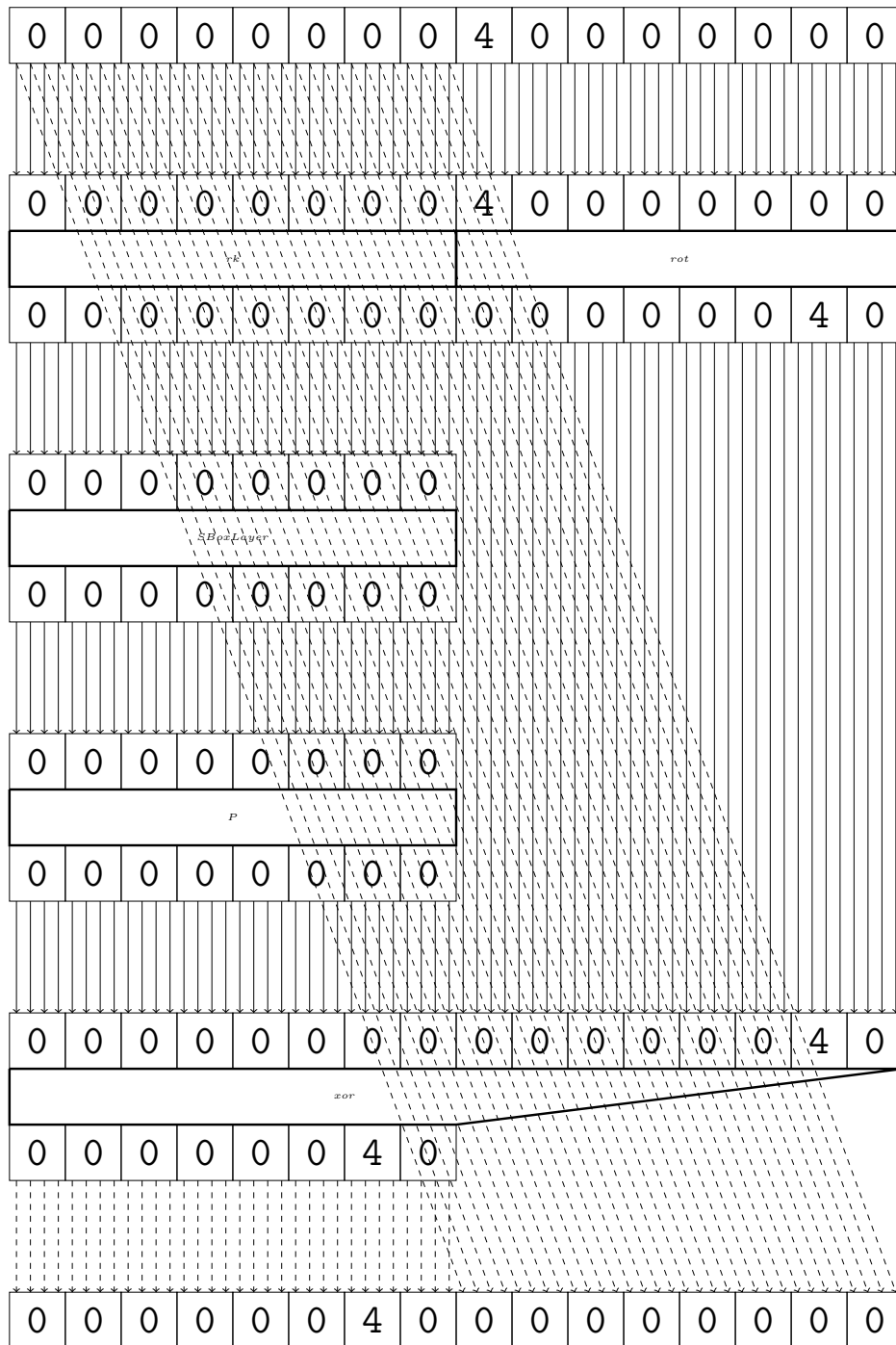
7 SBoxLayer

Maximal differential probability: $2^{-2.0}$



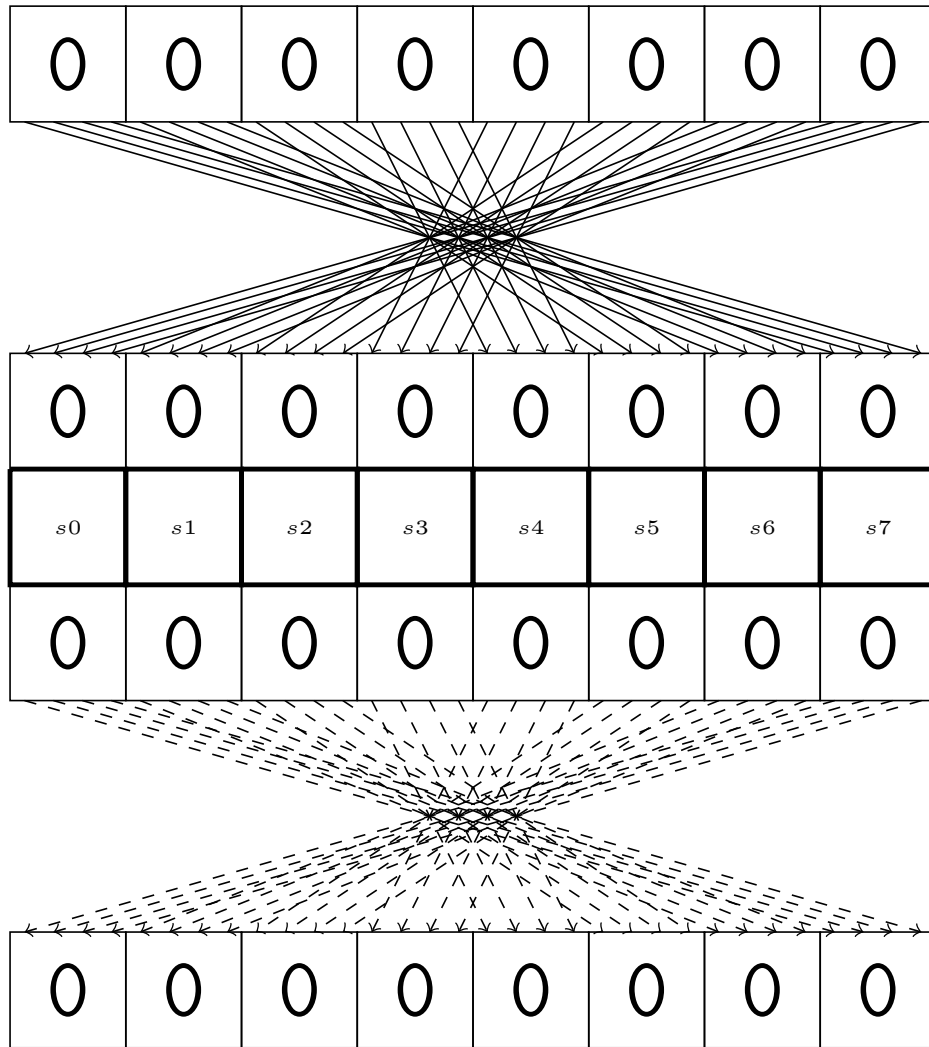
8 LBlock_round

Maximal differential probability: $2^{-0.0}$

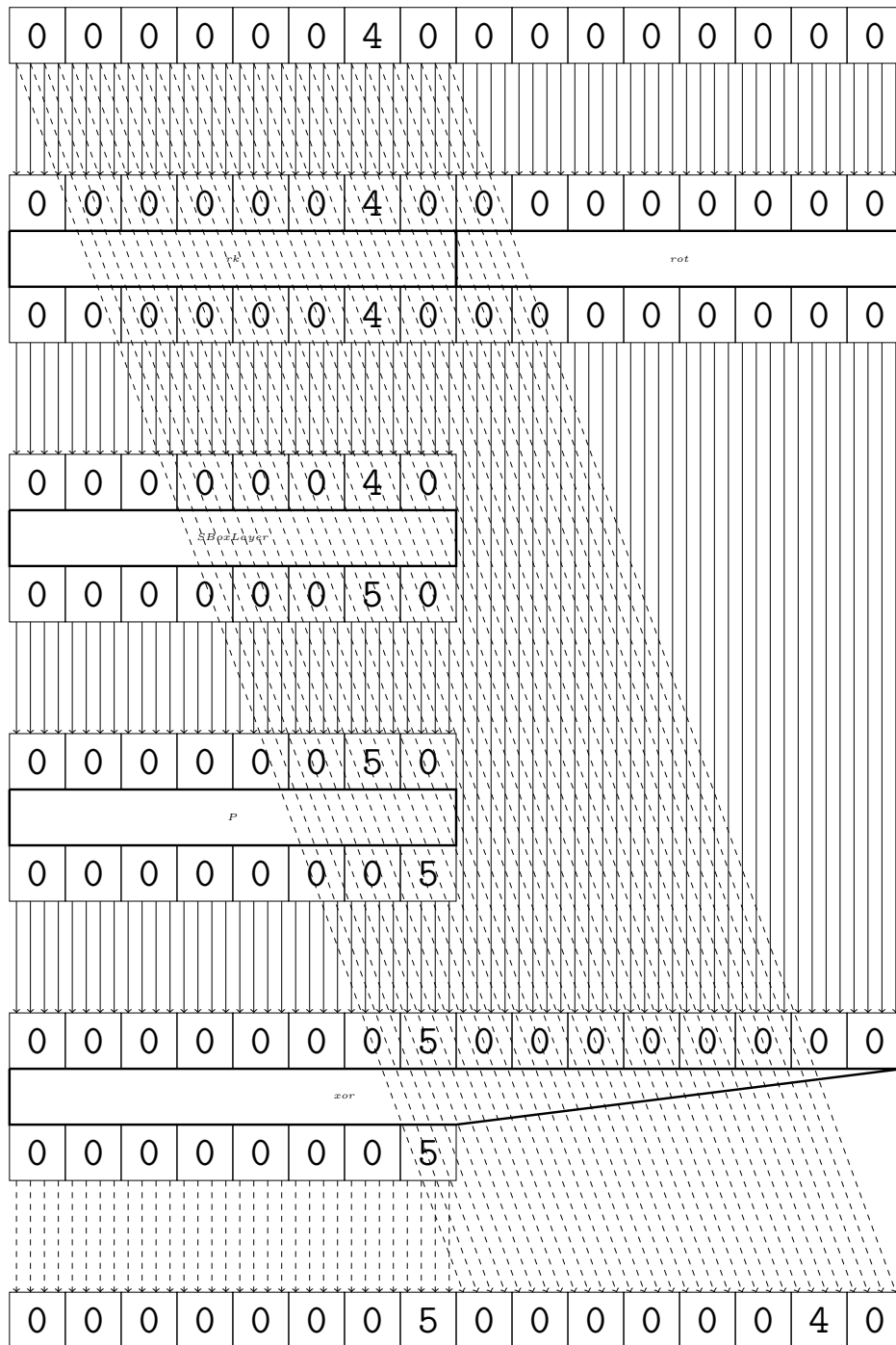


9 SBoxLayer

Maximal differential probability: $2^{-0.0}$

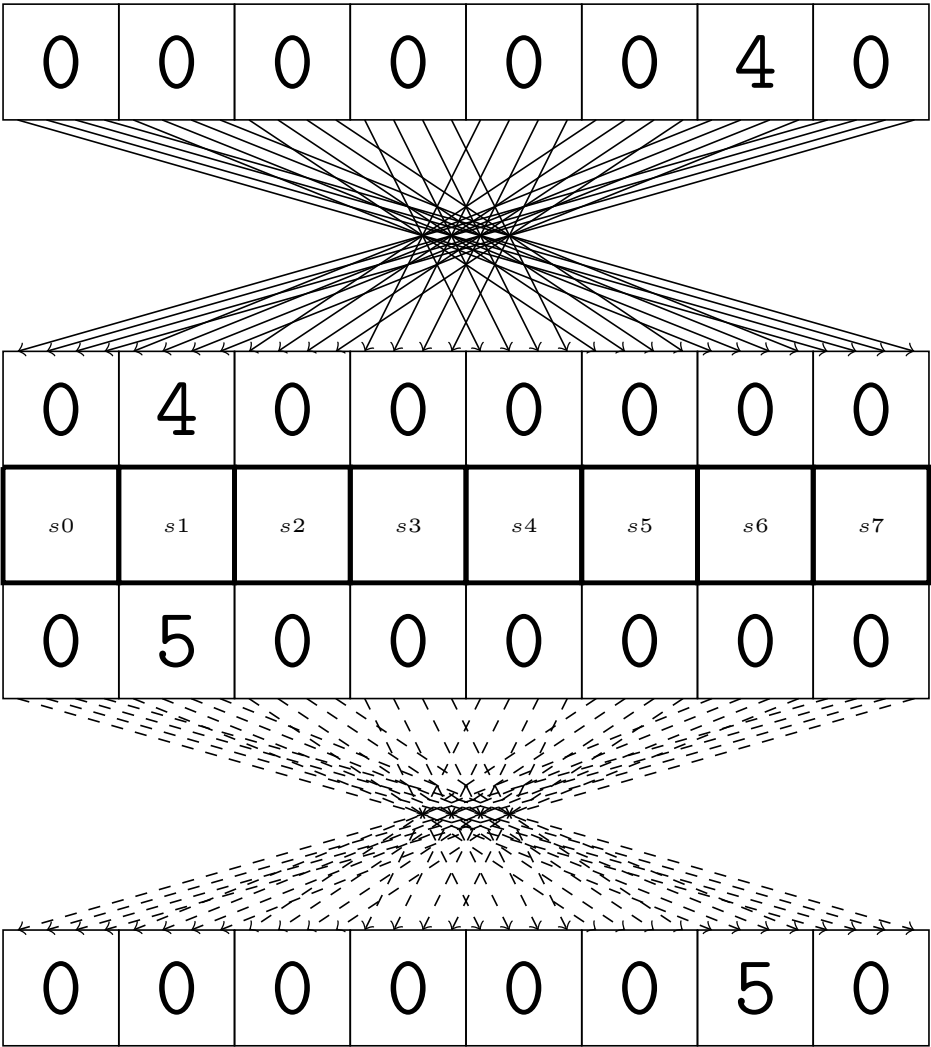


10 LBlock_round

Maximal differential probability: $2^{-2.0}$ 

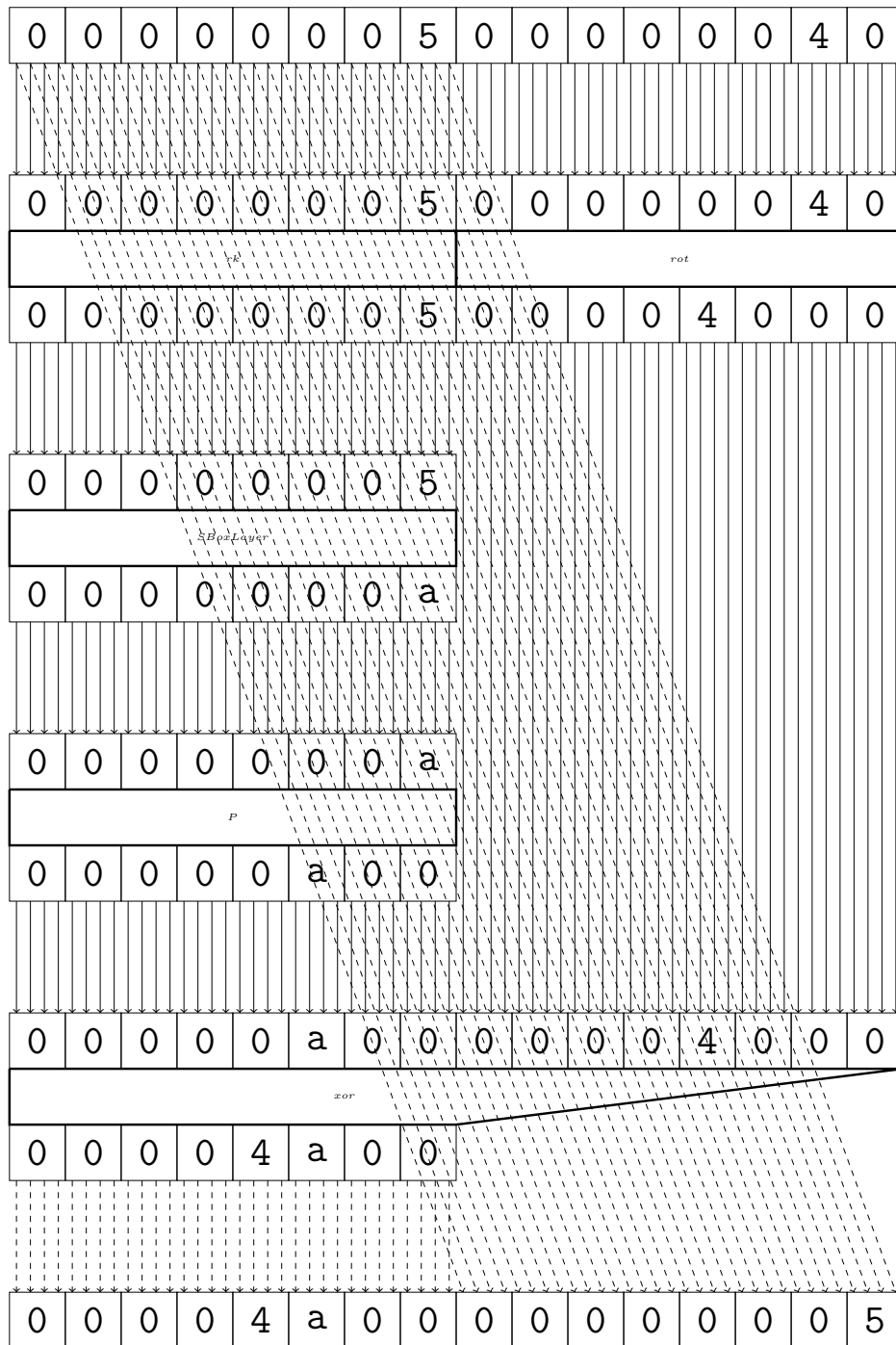
11 SBoxLayer

Maximal differential probability: $2^{-2.0}$



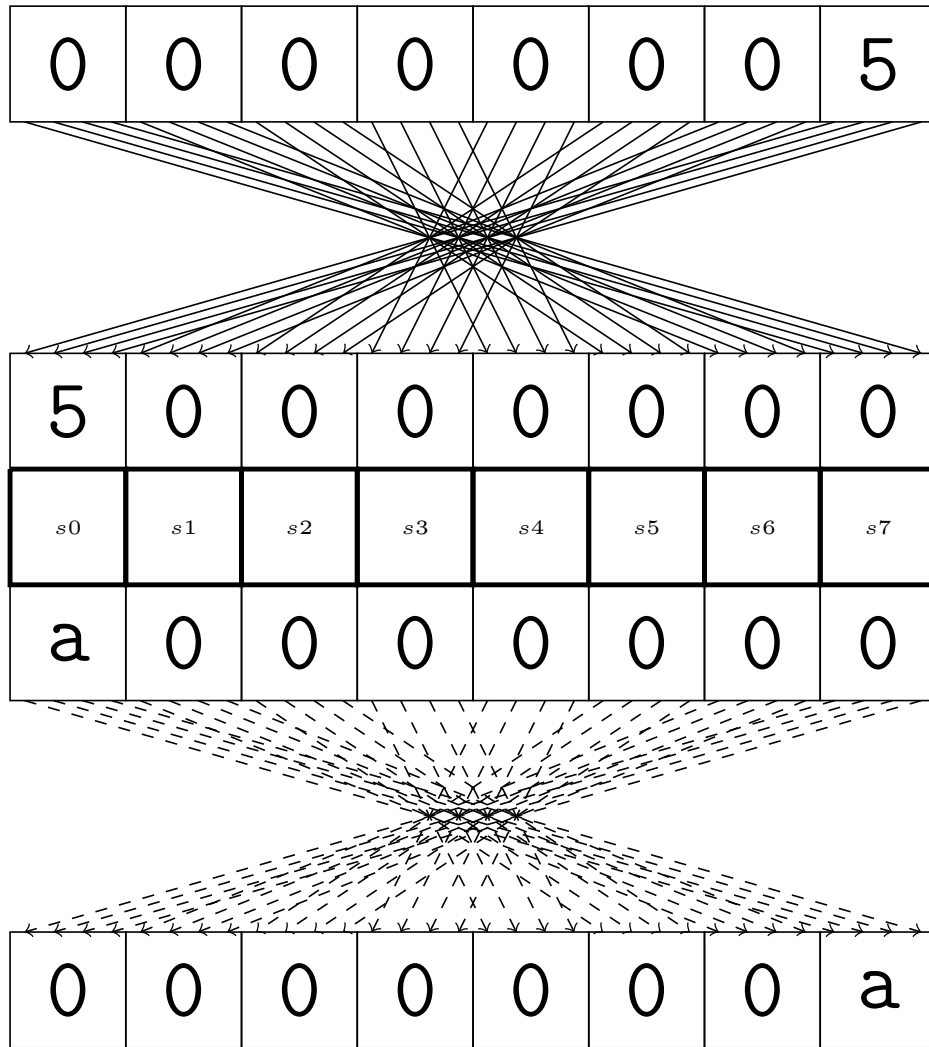
12 LBlock_round

Maximal differential probability: $2^{-2.0}$



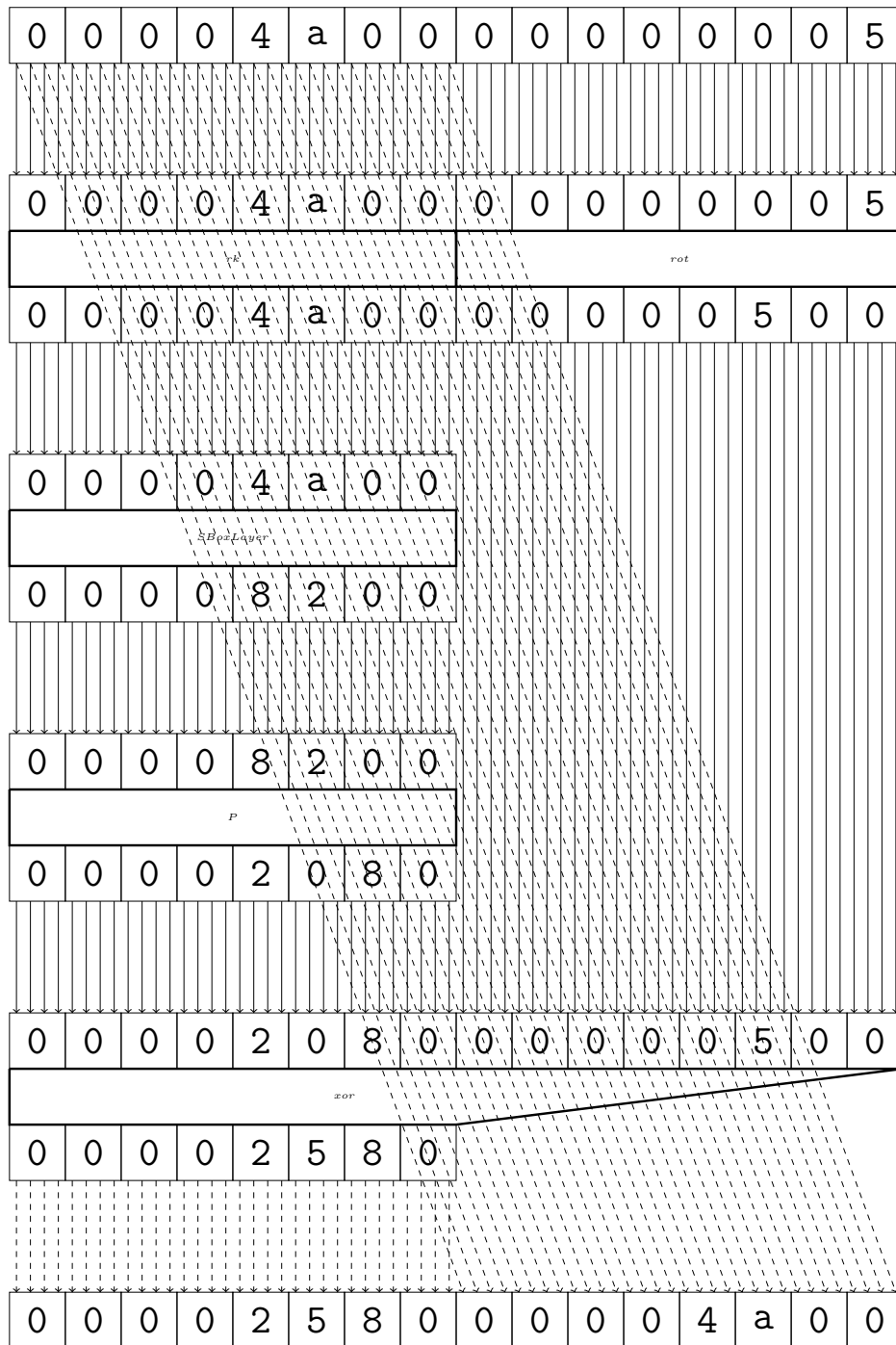
13 SBoxLayer

Maximal differential probability: $2^{-2.0}$



14 LBlock_round

Maximal differential probability: $2^{-4.0}$



15 SBoxLayer

Maximal differential probability: $2^{-4.0}$

